

The purpose of this database is to create a structured system that allows a local corner shop to streamline and improve its daily operations. To achieve this, the database includes five essential tables containing the key fields required for running the shop effectively.

The database will store the following information:

- Supplier data – supplier ID (PK), supplier name, address, phone number, email
- Inventory data – Product ID (PK), product name, product category, price, stock count, supplier ID(FK), Sale ID(FK).
- Sales data –Sales ID(PK), Sale date, quantity sold, total value, customer ID(FK), product ID(FK)
- Customer data – customer ID (PK), name, address, contact number, email, membership status, Sales ID(FK).
- Loyalty programme data – Membership_NO(PK), Customer_ID, points

The main users of this database will be:

- Staff – to check stock, record sales, and update loyalty points.
- Managers – to analyse sales trends, reorder stock, and maintain supplier relationships.

This database uses a relational schema where each table is linked through foreign keys to show how different parts of the shop's operations connect. Suppliers link to Inventory so each product can be traced to its source. Inventory links to Sales, allowing products to be associated with individual transactions. Sales then link to Customers, creating a relationship between purchases and the people who made them. Customers also link back to Sales, and the Loyalty table connects to Customers to track points. This schema is used because of the requirements of the store's tables and it allows for a non-technical user to easily understand the database for their own use.



After this we can start creating out database and tables. We will do this using the commands below:

```
CREATE DATABASE corner_shop;
```

Suppliers

```
CREATE TABLE Suppliers (  
    supplier_id INT PRIMARY KEY,  
    supplier_name VARCHAR(50),  
    address VARCHAR(100),  
    phone VARCHAR(20),  
    email VARCHAR(50)  
);
```

Inventory

```
CREATE TABLE Inventory (  
    product_id INT PRIMARY KEY,  
    product_name VARCHAR(50),  
    product_category VARCHAR(30),  
    price DECIMAL(5,2),  
    stock_count INT,  
    supplier_id INT,  
    FOREIGN KEY (supplier_id) REFERENCES Suppliers(supplier_id)  
);
```

Customers

```
CREATE TABLE Customers (  
    customer_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    address VARCHAR(100),  
    contact_number VARCHAR(20),  
    email VARCHAR(50),  
    membership_status VARCHAR(20)  
);
```

Loyalty

```
CREATE TABLE Loyalty (  
    membership_no INT PRIMARY KEY,  
    customer_id INT,  
    points INT,  
    FOREIGN KEY (customer_id) REFERENCES Customers(customer_id)  
);
```

Sales

```
CREATE TABLE Sales (  
    sale_id INT PRIMARY KEY,  
    sale_date DATE,  
    product_id INT,  
    customer_id INT,  
    quantity INT,  
    total_value DECIMAL(7,2),  
    FOREIGN KEY (product_id) REFERENCES Inventory(product_id),  
    FOREIGN KEY (customer_id) REFERENCES Customers(customer_id)  
);
```

After this we will need to populate the database with example data, we can do this using insert statements

Suppliers

```
INSERT INTO Suppliers  
VALUES (1, 'FreshFoods Ltd', 'London', '0201234567', 'info@freshfoods.com');
```

Inventory

```
INSERT INTO Inventory  
VALUES (101, 'Milk 1L', 'Dairy', 1.20, 40, 1);
```

Customers

```
INSERT INTO Customers  
VALUES (1, 'John Smith', '12 High Street', '07712345678', 'john@email.com', 'Active');
```

Loyalty

```
INSERT INTO Loyalty  
VALUES (5001, 1, 120);
```

Sales

```
INSERT INTO Sales  
VALUES (10001, '2024-05-10', 101, 1, 2, 2.40);
```

After inserting data, SELECT * statements are used to verify that everything is stored correctly.

Finally, before we hand over the database, we will focus on maintenance and strategies they can use to ensure the database remains consistent and accurate:

- Validate stock levels regularly
- Update customer details when needed
- Monitor loyalty points after each sale
- Use staff login access to prevent unauthorised edits
- Perform weekly backups to avoid data loss
- Secure the system with passwords and controlled access
- Review sales records for accuracy

To conclude, these maintenance practices ensure the database stays accurate, secure, and fully reliable for supporting the shop's daily operations.